

ABD

RFQ: Modular Workflow

Module 1: Detection

This module checks which email is relevant.

RFQ → Email Fetch → AI Filter → Tender Identification

(Emails arrive from Gmail or Outlook and AI filters them to determine if they are an RFQ or not.)

Module 2: Extraction

Once the RFQ is identified, the system extracts the necessary details.

Email Body → Metadata Extraction → Attachments → Folder Setup

(Client name, project reference, and all files are downloaded and secured in the database and folders.)

Module 3: Drafting

As you noted, this is where user control comes in.

Draft Generation → User Edit → AI Enhance → Final Review

(The system generates a draft, which you can edit or ask the AI to "enhance" further.)

Module 4: Execution

The final stage where the task is completed.

Draft Finalized → Gmail/Outlook Send → Sent Status Update → Dashboard Update

(When you click 'Send', the system sends the actual email and updates the report on the dashboard.)

RFQ → Email → Draft → User Edit/Enhance → We Can Send ☒

Rehan

1. Dashboard

Frontend: This is your main control centre where a snapshot of the entire business is visible.

[View Stats](#) → [Monitor Recent Orders](#) → [Check System Health](#)

(If you need to check total sent emails or total vendors, the dashboard shows updated data immediately.)

2. Email Inbox & Scanner

Frontend & Backend: This module finds relevant emails from your inbox and brings them into the system.

[Fetch Emails](#) → [Keyword Filtering \(RFQ/Urgent\)](#) → [Email List Display](#)

(If an email is not related to procurement, the system automatically "Ignores" it.)

3. AI Extraction Engine

Backend: This is the "Brain" of the system that extracts products and quantities from complex email text.

[Parse Email Text](#) → [LLM Analysis](#) → [Structured Product List](#)

(If there are 10 different types of items in an email, the system creates a separate record for each with "Product Name" and "Qty".)

4. Supplier Database & Matcher

Frontend & Backend: This module manages your approved vendors and links them to products.

[Manage Vendor List](#) → [Category Mapping \(Trade Wise\)](#) → [Propose Matching Suppliers](#)

(If the extractor identifies "Cement," the Matcher module immediately maps it to vendors in the "Builder" category.)

5. RFQ Draft Builder

Frontend & Backend: This module prepares customized request emails for each vendor.

[Select Vendors](#) → [Auto-Generate Drafts](#) → [Manual Review / AI Improvement](#)

(You can review the draft and give the AI instructions like "Make this more professional," and the system will rewrite it.)

6. Smart File Processor

Backend & Frontend: This module scans PDFs, Excel files, and images to extract their data.

Upload Document → Multi-format Parsing (OCR/Excel) → Content Injection for AI

(If you upload a vendor's PDF quote, the system reads it and extracts the product and rate data.)

7. Price Comparison Table

Frontend & Backend: This module normalizes rates from all vendors to show an "Apple-to-Apple" comparison.

Input Vendor Quotes → Normalize Prices → Highlight BEST vs HIGH rates

(The vendor offering the cheapest rate will be automatically highlighted at the top with a "Recommended" badge.)

8. AI Assistant Chatbot

Frontend Interface: This is an interactive bot that assists you with any task in the system.

User Question → Context Search → Natural Language Response

(If you ask, "How to add a supplier?", the bot will immediately give you the step-by-step procedure.)

Usman

Module 1: Orchestrator (Pipeline Controller)

This is the "boss" module that runs everything in the correct sequence.

Tender Folder Detected → Read All Files (01-07) → Run Each Module One by One → Update Status

(When files are ready, the Orchestrator calls each module. If one fails, it logs the error and moves to the next.)

Module 2: Instruction Extractor

Extracts fundamental tender instructions like deadlines, formats, and penalties.

01_Instructions/ Files → AI Read → Extract Deadlines + Submission Rules + Contacts + Penalties

(It identifies rules like "Submit by Jan 30 in hardcopy" with document page numbers and confidence scores.)

Module 3: Scope Extractor

Identifies exactly what work needs to be done and what is excluded.

02_Scope_of_Work/ Files → AI Read → Work Packages + Deliverables + Exclusions + Assumptions

(Creates a WBS like "Civil, MEP, Fire Safety." Every line is traced to its source document; unclear items are flagged.)

Module 4: Drawing Analyzer

Scans drawings to identify disciplines, revisions, and sheet numbers. **03_Drawings/ PDFs → Convert to Images → AI Vision Analysis → Drawing Register (Type/Revision/Sheet)**

(Uses AI Vision to determine if a drawing is Architectural or MEP and links it directly to specific scope items.)

Module 5: BOQ Parser

Extracts items, quantities, and units from the Bill of Quantities. **05_BOQ/ Files (Excel/PDF) → Parse Tables → Extract Items + Quantities + Units → Map to Scope + Drawings**

(Uses specialized libraries to read tables. It never changes quantities but reports mismatches between the BOQ and scope.)

Module 6: Compliance Engine (Saudi Standards Check)

Checks the tender requirements against specific Saudi Arabian standards. **Knowledge Base (SBC/SASO/Aramco) → Detect Applicable Standards → Check Evidence → Compliance Matrix**

(Automatically applies standards like Aramco SAES if the client is Aramco. Results are marked as Compliant, Gap, or Non-Compliant.)

Module 7: Risk Detector

Identifies errors, conflicts, or missing information across the entire tender.

ALL Previous Results → AI Scan → Missing Docs + Conflicts + BOQ Gaps + Timeline Risks

(Dependent on all previous modules. If a drawing is referenced in the scope but missing from the register, it flags a High Risk.)

Module 8: Assumption Generator

Creates draft assumptions and exclusions based on identified gaps.

Risks + Scope Gaps → AI Draft → Assumptions + Exclusions (All marked DRAFT)

(If details are unclear, the system drafts statements like "We assume site access by Week 2" for human approval.)

Module 9: RFI Generator

Prepares professional drafts of questions (Requests for Information) to ask the client.

Risks + Compliance Gaps → AI Draft → RFI Questions with Source References

(Drafts questions citing exact document sections. These remain as drafts until a human reviews and approves them.)

Module 10: Readiness Scorer

Calculates how ready the tender is for submission as a percentage.

ALL Tables Count → Calculate Completeness → Score % + Risk Level + Breakdown

(Weights categories like Instructions, Scope, and BOQ. Results like "82% Ready, Medium Risk" guide the final decision.)

Module 11: Handover Builder

Creates the final approved data package for the procurement team. **Only APPROVED Items → Build JSON Package → Save to DB + 08_Output/handover.json**

(Only items approved by a human are included. This clean data package is then handed over to the Procurement Agent.)

